

NETWORKED CONTROL

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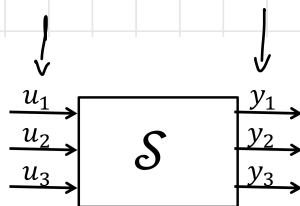
PART II: DECENTRALIZED

AND DISTRIBUTED
CONTROL STRUCTURES

The present notes are not to be considered as a substitute of the personal notes or of the official lecture notes freely available on webeep.

1. MODELS

$$\mathcal{S}: \begin{cases} \dot{x} = Ax + Bu \\ y = Cx \end{cases}$$



To "DECOMPOSE MODELS";

I). Define N subsystems $\mathcal{S}_i$ $i=1, \dots, N$

• DECOMPOSE VECTORS u, y

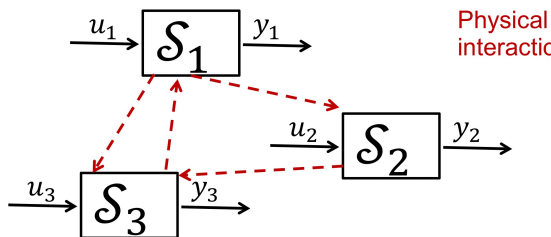
$$u = \begin{bmatrix} u_1 \\ \vdots \\ u_N \end{bmatrix}, \quad y = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix}$$

$$u: \in \mathbb{R}^{u_i}; \quad \sum u_i = u$$

$$y: \in \mathbb{R}^{p_i}; \quad \sum p_i = p$$

IT CAN BE DONE BY:

- PHYSICAL INSPECTION
- MATHEMATICAL TOOLS
(E.G. RELATIVE GAIN ARRAY RGA)



II) DECOMPOSE THE MATRICES B AND C

$$B = [B_1 \ B_2 \ \dots \ B_N]$$

$$\dot{x} = Ax + B \begin{bmatrix} u_1 \\ \vdots \\ u_N \end{bmatrix} = Ax + [B_1 \ \dots \ B_N] \begin{bmatrix} u_1 \\ \vdots \\ u_N \end{bmatrix} = Ax + B_1 u_1 + \dots + B_N u_N$$

EXAMPLE: $u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$

$$\Rightarrow Bu = B \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \overbrace{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}^{B_1} u_1 + \overbrace{\begin{bmatrix} 2 \\ 3 \end{bmatrix}}^{B_2} u_2$$

$$y = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix} = C x = \begin{bmatrix} c_1 \\ \vdots \\ c_N \end{bmatrix} x$$

$\swarrow$
 $\begin{bmatrix} c_1 \\ \vdots \\ c_N \end{bmatrix}$

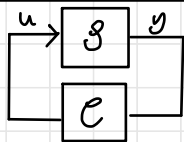
$\Rightarrow$ For all $i=1, \dots, N$

$$\Rightarrow \begin{cases} y_i = c_i x \\ \dot{x} = A x + \sum_{j=1}^N B_j u_j \end{cases}$$

$(u_i, y_i) \Rightarrow i$ -TH channel

2. CONTROL SCHEMES AND CONTROLLABILITY PROPERTIES

A) CENTRALIZED CONTROL SCHEMES



$$p: \begin{cases} \dot{x} = Ax + Bu \\ y = Cx \end{cases}$$

1) Output-feedback static control

$$u = K_y y$$

$$\Rightarrow \dot{x} = Ax + BK_y Cx = \underbrace{(A + BK_y C)}_{\text{CLOSED-LOOP SYSTEM MATRIX}} x$$

2) State-feedback static control

$$u = K_x x \quad (y = x \Rightarrow C = I)$$

$$\Rightarrow \dot{x} = (A + BK_x) x$$

(A, B) CONTROLLABLE $\Rightarrow \exists K_x$ s.t. all eigenvalues of $A + BK_x$ can be placed at will

3) Output-feedback dynamic controller

e is a dynamic system, i.e.,

$$e: \begin{cases} \dot{\bar{x}} = A_c \bar{x} + B_c y \\ u = K_x \bar{x} + K_y y \end{cases}$$

$\rightarrow$ e.g. REALIZATION OF A PID,
OR STATE ESTIMATOR (OBSERVER) +
CONTROLLER (*)

②

$\bar{x}$: STATE ESTIMATES

OBSERVER $\left\{ \begin{array}{l} \dot{\bar{x}} = A\bar{x} + Bu + L(y - C\bar{x}) \\ \text{CONTROLLER} \quad u = K_x \bar{x} \end{array} \right.$

$\xrightarrow{\quad} \text{output estimate}$

$$\begin{aligned} \dot{\bar{x}} &= A\bar{x} + BK_x \bar{x} + Ly - LC\bar{x} \\ &= \underbrace{(A + BK_x - LC)}_{A_c} \bar{x} + \underbrace{Ly}_{B_c} \end{aligned}$$

TO STUDY ITS DYNAMICS WE DEFINE

$e = x - \bar{x}$ AND WE DERIVE:

$$\dot{e} = \dot{x} - \dot{\bar{x}} = Ax + \cancel{Bu} - (A\bar{x} + \cancel{Bu} + \underbrace{LC(x - \bar{x})}_{= LCe})$$

$$= Ax - A\bar{x} - LCe = Ae - LCe$$

$$= (A - LC)e \Rightarrow \text{YOU CAN PLACE THE EIGENVALUES OF } A - LC \text{ IFF } (A, C) \text{ IS OBSERVABLE}$$

DYNAMICS OF THE CONTROL SYSTEM:

$$\begin{cases} \dot{e} = (A - LC)e \\ \dot{\bar{x}} = A\bar{x} + BK_x \bar{x} = A\bar{x} + BK_x x - BK_x e \\ \quad = (A + BK_x)x - BK_x e \end{cases}$$

$$\Rightarrow \begin{bmatrix} \dot{e} \\ \dot{\bar{x}} \end{bmatrix} = \begin{bmatrix} A - LC & 0 \\ -BK_x & A + BK_x \end{bmatrix} \begin{bmatrix} e \\ \bar{x} \end{bmatrix} \Rightarrow \text{CONTROLLABILITY OF } (A, B) \text{ ALLOWS TO DEFINE } K_x \text{ SUCH THAT } A + BK_x \text{ IS HURWITZ}$$

• OBSERVABILITY OF (A, C) ALLOWS TO DEFINE L SUCH THAT $A - LC$ IS HURWITZ

4) Controllability and observability

- A LINEAR TIME-INVARIANT SYSTEM

$$\dot{x} = Ax + Bu$$

IS CONTROLLABLE IFF $\forall x(0) \exists u(t), t \in [0, T)$ SUCH THAT $x(T) = 0$

- A LINEAR TIME-INVARIANT SYSTEM

$$\begin{cases} \dot{x} = Ax \\ y = Cx \end{cases}$$

IS OBSERVABLE IFF WE CAN RECONSTRUCT $x(0)$ BY KNOWING $y(t), t \in [0, T]$

- CONTROLLABILITY TESTS

THE SYSTEM (A, B) IS CONTROLLABLE IFF ONE OF THE FOLLOWING EQUIVALENT PROPERTIES HOLD:

I) $\text{rank}(S_c) = n$ (n : ORDER OF THE SYSTEM)

$$S_c = \begin{bmatrix} B & AB & A^2B & \dots & A^{n-1}B \end{bmatrix} \quad \text{CONTROLLABILITY MATRIX}$$

II) PBH TEST (Popov - Belevich - Hautus)

$$\text{rank}[\lambda I - A \quad B] = n \quad \forall \lambda \in \mathbb{C} \quad (\text{i.e., for all } \lambda = \lambda_i \text{ EIGENVALUES OF } A)$$

IN FACT $\lambda I - A$ IS SINGULAR ($\det(\lambda I - A) = 0$, i.e. $\text{rank}(\lambda I - A) < n$) ONLY FOR $\lambda = \lambda_i$, i.e., IN THE EIGENVALUES!

⇒ IMPORTANT! EIGENVALUE λ_i CAN BE MOVED IFF

$$\text{rank}[\lambda_i I - A \quad B] = n, \text{ i.e.}$$

IF $\text{rank}[\lambda_i I - A \quad B] < n$ THE MODE CORRESPONDING WITH λ_i CANNOT BE "CONTROLLED"

• OBSERVABILITY TESTS

(A, c) IS OBSERVABLE IFF ONE OF THE FOLLOWING EQUIVALENT CONDITIONS HOLD.

• $\text{rank}(S_o) = n$, $S_o = \begin{bmatrix} c \\ cA \\ \vdots \\ cA^{n-1} \end{bmatrix}$ OBSERVABILITY MATRIX

• PBH TEST

$$\text{rank} \begin{bmatrix} \lambda I - A \\ c \end{bmatrix} = n \quad \forall \lambda \in \mathbb{C}$$

$\Downarrow$
 $\forall \lambda = \lambda_i$ EIGENVALUES OF A

$\Rightarrow$ IF $\text{rank} \begin{bmatrix} \lambda_i I - A \\ c \end{bmatrix} = n \rightarrow \lambda_i$ IS AN OBSERVABLE MODE

$\text{rank} \begin{bmatrix} \lambda_i I - A \\ c \end{bmatrix} < n \Rightarrow \lambda_i$ IS NOT OBSERVABLE

MODES THAT ARE

- NOT CONTROLLABLE AND/OR
- NOT OBSERVABLE

ARE FIXED MODES

To define fixed modes consider the output feedback static control law:

$$u = K_y y \quad \Rightarrow \quad \dot{x} = (A + BK_y C)x$$

DEFINITION:

$$\Lambda_{CFM} = \bigcap_{K_y \in \mathbb{R}^{m \times p}} \text{spectrum}(A + BK_y C)$$

$\rightarrow$ set of all eigenvalues of a matrix

IS THE SET OF CENTRALIZED FIXED MODES (CFM)

Remarks

- IF λ_i IS NOT FIXED: • IT CANNOT BE MOVED IN DESIRED POSITIONS IN GENERAL USING AN OUTPUT FEEDBACK STATIC CONTROL LAW

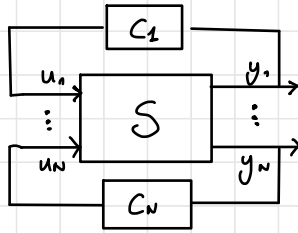
! BUT !

- YOU CAN MOVE IT AT WILL USING A DYNAMIC OUTPUT FEEDBACK CONTROLLER
- IF λ_i IS FIXED IT CANNOT BE MOVED WITH ANY TYPE OF CONTROLLER $\Rightarrow$ IT IS STRUCTURALLY FIXED
- IF λ_i IS EIGENVALUE OF $A \Rightarrow e^{\lambda_i h}$ IS EIGENVALUE OF $\bar{F} = e^{A h}$
- IF λ_i IS A C.F.M. FOR THE CONTINUOUS-TIME SYSTEM $\Rightarrow e^{\lambda_i h}$ IS A CFM FOR THE DISCRETE-TIME SYSTEM

✓

$$\lambda_{\text{CFM}}^{\text{DISCRETE-TIME}} = \bigcap_{K_y \in \mathbb{R}^{m \times p}} \text{spectrum} (F + G K_y H)$$

B. DECENTRALIZED CONTROL



1) Static output feedback control law

$$u_i = K_y^i y_i \quad \text{FOR ALL } i = 1, \dots, N$$

$$u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_N \end{bmatrix} = \underbrace{\begin{bmatrix} K_y^1 & 0 & \dots & 0 \\ 0 & K_y^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & K_y^N \end{bmatrix}}_{K_y^d} \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}}_y = K_y^d y$$

THE GAIN IS
BLOCK-DIAGONAL, I.E.

$$\dot{x} = Ax + Bu$$

$$\begin{aligned} &= Ax + B K_y^d y \\ &= (A + B K_y^d C) x \end{aligned}$$

$$K_y^d \in \mathbb{R}^d$$

set of all block-diagonal
matrices of suitable
dimensions

2) Dyn > moc output feedback controllers

C_i IS A DYNAMICAL SYSTEM WITH STATE $\bar{x}_i$, FOR ALL $i = 1, \dots, N$

$$\begin{cases} \dot{\bar{x}}_i = \bar{A}_i \bar{x}_i + \bar{B}_i y_i \\ u_i = \bar{K}_x^i \bar{x}_i + \bar{K}_y^i y_i \end{cases}$$

REMARK: THE STATE OF S (I.E., x)
MAY NOT BE OBSERVABLE FROM JUST y_i

$\Rightarrow$ COMMONLY $\bar{x}_i$ IS NOT THE
ESTIMATE OF x

E.G., REALIZATION OF LOCAL PI
CONTROLLERS

3) Static state-feedback control

IN THE "CENTRALIZED" DOMAIN ~~$u_i = K_i x$~~ $\Rightarrow$ IN THE DECENTRALIZED FRAMEWORK WE CANNOT ASSUME THAT x IS KNOWN BY ALL LOCAL CONTROLLERS

I) PARTITION ALSO THE STATE VECTOR x INTO NON OVERLAPPING SUBVECTORS $x_i, i=1, \dots, N$

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_N \end{bmatrix}$$

$$[0 \dots 0 I 0 \dots 0]$$

II) STATE FEEDBACK :

$$u_i = K_x^i \boxed{x_i} = K_x^i C_i x$$

$$u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_N \end{bmatrix} = \underbrace{\begin{bmatrix} K_x^1 & 0 & \dots & 0 \\ 0 & K_x^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & K_x^N \end{bmatrix}}_{K_x^d \in K^d} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix} = K_x^d x$$

$$K_x^d \in K^d$$

$$\Rightarrow \dot{\bar{x}} = (A + B K_y^d) x$$

In reality, it must be considered as an output-feedback (static) controller, i.e. $x_i = C_i x$ FOR ALL i

4) Decentralized fixed modes

DEFINITION :

$$\Lambda_{DFM} = \bigcap_{K_y^d \in K^d} \text{spectrum}(A + B K_y^d C)$$

IS THE SET OF DECENTRALIZED FIXED MODES (DFM)

Remark: recall that $\Lambda_{CFM} = \bigcap_{K_y \in \mathbb{R}^{m \times p}} \text{spectrum}(A + B K_y C)$

and note that $K^d \subset \mathbb{R}^{m \times p}$

THEREFORE: $\lambda_i \in \Lambda_{CFM} \Rightarrow \lambda_i \in \Lambda_{DFM}$

~~$\Leftarrow$~~

THIS IMPLIES THAT $\Lambda_{CFM} \subseteq \Lambda_{DFM}$

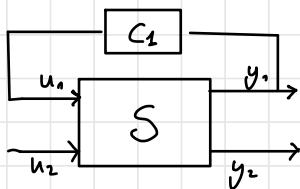
Theorem:

- IF THE SYSTEM HAS NO DECENTRALIZED FIXED MODES
⇒ ALL EIGENVALUES CAN BE PLACED IN ARBITRARY DESIRED POSITIONS USING A DECENTRALIZED (LINEAR, TIME-INVARIANT, FINITE-DIMENSIONAL) DYNAMIC OUTPUT FEEDBACK CONTROL LAW
- WITH A LINEAR, TIME-INVARIANT, FINITE-DIMENSIONAL DECENTRALIZED CONTROLLER WE CANNOT MOVE DFMS

↓
No delays

NECESSARY CONDITION FOR DECENTRALIZED FIXED MODES

EXAMPLE:



THIS (DECENTRALIZED) STRUCTURE IS SUFFICIENT IF, AT THE SAME TIME,

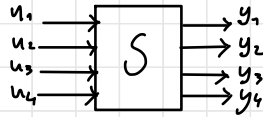
- (A, B_1) IS CONTROLLABLE
- (A, C_1) IS OBSERVABLE

IN GENERAL, A MODE λ CAN BE FIXED ONLY IF (NECESSARY CONDITION) FOR ALL CHANNELS (u_i, y_i) , $i=1, \dots, N$:

- λ IS NOT OBSERVABLE BY y_i
(i.e., $\text{rank} \begin{bmatrix} \lambda I - A \\ C_i \end{bmatrix} < n$) , AND/OR
- λ IS NOT CONTROLLABLE BY u_i
(i.e., $\text{rank} [\lambda I - A \quad B_i] < n$)

EXAMPLE: CONSIDER A SYSTEM WITH 4 CHANNELS

ASSUME, FOR EXAMPLE THAT, FOR A GIVEN EIGENVALUE $\bar{\lambda}$ OF A :



- 1) $\text{rank} [\bar{\lambda}I - A, B] < n$, $\text{rank} \begin{bmatrix} \bar{\lambda}I - A \\ C \end{bmatrix} = n \Rightarrow \text{not cont.}$
- 2) $\text{rank} [\bar{\lambda}I - A, B] < n$, $\text{rank} \begin{bmatrix} \bar{\lambda}I - A \\ C \end{bmatrix} < n \Rightarrow \text{not cont, not obs}$
- 3) $\text{rank} [\bar{\lambda}I - A, B] = n$, $\text{rank} \begin{bmatrix} \bar{\lambda}I - A \\ C \end{bmatrix} < n \Rightarrow \text{not obs}$
- 4) $\text{rank} [\bar{\lambda}I - A, B] < n$, $\text{rank} \begin{bmatrix} \bar{\lambda}I - A \\ C \end{bmatrix} < n \Rightarrow \text{not cont, not obs}$

Define the (channel) index set $\mathcal{I} = \{1, \dots, N\}$. In the example

$$\mathcal{I} = \{1, 2, 3, 4\}$$

We define a disjoint partition of $\mathcal{I}$ into sets $\mathcal{D}$ and $\mathcal{K}$ such that



$$\begin{aligned} \mathcal{D} \cup \mathcal{K} &= \mathcal{I} \\ \mathcal{D} \cap \mathcal{K} &= \emptyset \end{aligned}$$

INDEXES OF CHANNELS
WHERE $\bar{\lambda}$ IS NOT
CONTROLLABLE

INDEXES OF
CHANNELS SUCH THAT
 $\bar{\lambda}$ IS NOT OBSERVABLE

IN THE EXAMPLE $\exists$ DIFFERENT POSSIBILITIES:

- I) $\mathcal{D} = \{1, 2, 4\}$, $\mathcal{K} = \{3\}$
- II) $\mathcal{D} = \{1, 4\}$, $\mathcal{K} = \{2, 3\}$
- III) $\mathcal{D} = \{1, 2\}$, $\mathcal{K} = \{3, 4\}$
- IV) $\mathcal{D} = \{1\}$, $\mathcal{K} = \{2, 3, 4\}$

IN GENERAL WE HAVE:

$$\mathcal{D} = \{i_1, \dots, i_k\} \quad (k \text{ indices})$$

$$\mathcal{K} = \{i_{k+1}, \dots, i_N\} \quad (N-k \text{ indices})$$

THE PREVIOUS NECESSARY CONDITION CAN BE REWRITTEN AS FOLLOWS.

A MODE λ CAN BE FIXED ONLY IF
(NECESSARY CONDITION) $\exists$ DISJOINT PARTITION OF $\mathcal{I}$ INTO $\mathcal{D}$ and $\mathcal{H}$ such that

$$\text{FOR ALL } i \in \mathcal{D}, \text{rank}[\lambda I - A \quad B_i] < n$$

$$\text{FOR ALL } i \in \mathcal{H}, \text{rank} \begin{bmatrix} \lambda I - A \\ C_i \end{bmatrix} < n$$

A STRONGER CONDITION CAN BE DERIVED DEFINING:

$$B_D = [B_{i_1}, B_{i_2}, \dots, B_{i_k}]$$

$$C_H = \begin{bmatrix} C_{i_{k+1}} \\ \vdots \\ C_{i_n} \end{bmatrix}$$

A MODE λ CAN BE FIXED ONLY IF
(NECESSARY CONDITION) $\exists$ DISJOINT PARTITION OF $\mathcal{I}$ INTO $\mathcal{D}$ and $\mathcal{H}$ such that

$$\text{rank}[\lambda I - A \quad B_D] < n, \text{rank} \begin{bmatrix} \lambda I - A \\ C_H \end{bmatrix} < n \quad (*)$$

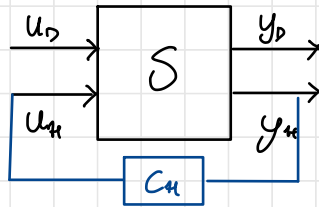
IF WE DEFINE :

$$u_D = \begin{bmatrix} u_{i_1} \\ \vdots \\ u_{i_n} \end{bmatrix} \quad y_D = \begin{bmatrix} y_{i_1} \\ \vdots \\ y_{i_n} \end{bmatrix}$$

$$u_H = \begin{bmatrix} u_{i_{n+1}} \\ \vdots \\ u_{i_N} \end{bmatrix} \quad y_H = \begin{bmatrix} y_{i_{n+1}} \\ \vdots \\ y_{i_N} \end{bmatrix}$$

WE CAN REGARD THE SYSTEM AS IF IT HAD 2 MAIN CHANNELS :

- THE UNCONTROLLABLE CHANNEL (u_D, y_D)
- THE UNOBSERVABLE CHANNEL (u_H, y_H)



λ CAN BE A DECENTRALIZED FIXED MODE ONLY IF

- λ IS NOT CONTROLLABLE BY u_D
 - λ IS NOT OBSERVABLE BY y_H
- } CONDITION (*)



THIS CONDITION IS NECESSARY.
IS IT ALSO SUFFICIENT?

EXAMPLE

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u_1 + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u_2$$

$$y_1 = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$y_2 = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \Rightarrow S_0 = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} = \text{rank}(S_0) = n$$

$$C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$S_c = \begin{bmatrix} B & AB \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

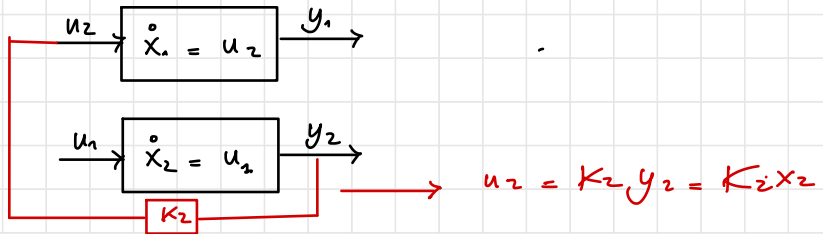
THE EQUATIONS CAN BE REWRITTEN AS

$$\hookrightarrow \text{rank}(S_c) = 2$$

$$\begin{aligned} \dot{x}_1 &= u_2 \\ \dot{x}_2 &= u_1 \end{aligned}$$

$$\begin{aligned} y_1 &= x_1 \\ y_2 &= x_2 \end{aligned} \Rightarrow \text{DECOUPLED SYSTEMS}$$

$\Downarrow$
NO CFM (CENTRALIZED FIXED MODES)



WE ANALYZE MODE x_1 ($\lambda = 0$) $\Rightarrow D = \{1\}$
 $\mathcal{H} = \{2\}$

When the loop on channel (u_2, y_2) is closed with $u_2 = K_2 y_2$, the system becomes

$$\begin{aligned} \dot{\hat{x}} &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \hat{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u_1 + \begin{bmatrix} 1 \\ 0 \end{bmatrix} K_2 \begin{bmatrix} 0 & 1 \end{bmatrix} \hat{x} \\ &= \underbrace{\begin{bmatrix} 0 & K_2 \\ 0 & 0 \end{bmatrix}}_{\hat{A}} \hat{x} + \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{B_1} u_1 \end{aligned}$$

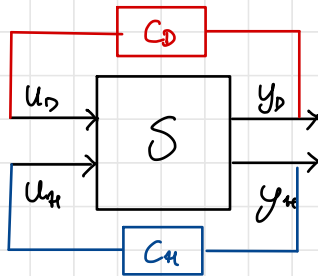
THE CONTROLLABILITY MATRIX OF THE PAIR

$(\tilde{A}, B_1)$ IS

$$S_c = \begin{bmatrix} 0 & k_2 \\ 1 & 0 \end{bmatrix} \rightarrow \text{rank}(S_c) = 2 \quad \text{if } k_2 \neq 0$$

$\downarrow$ $\downarrow$
 B_1 $\tilde{A}B_1$

$\Rightarrow$ THIS EXAMPLE HIGHLIGHTED THAT CONDITION (*) IS NOT "COMPLETE".
IN FACT, CONSIDERING AGAIN THE SYSTEM



λ CAN BE A DECENTRALIZED FIXED MODE ONLY IF

- λ IS NOT CONTROLLABLE BY u_D WHEN THE LOOP ON CHANNEL H IS BOTH OPEN AND CLOSED BY

$$u_H = \text{diag}(k_{H1}, \dots, k_{Hn}) y_H$$

- λ IS NOT OBSERVABLE BY y_D WHEN THE LOOP ON CHANNEL D IS BOTH OPEN AND CLOSED BY

$$u_D = \text{diag}(k_{D1}, \dots, k_{Dn}) y_D$$

CONDITION
(**)

IT IS POSSIBLE TO PROVE THAT CONDITION (**) IS BOTH NECESSARY AND SUFFICIENT, AND THAT IT CAN BE REFORMULATED AS FOLLOWS:

IF A DISJOINT PARTITION OF S INTO D AND H SUCH THAT

$$\text{rank} \begin{bmatrix} \lambda I - A & B_D \\ C_H & 0 \end{bmatrix} < n \quad (i)$$

To check (i) we need to

- 1) Consider each λ (mode) separately
- 2) Define all the "candidate" partitions $\mathcal{D}$ and $\mathcal{K}$
- 3) Check (i)

↓
Cumbersome for large-scale systems

"EASY" NUMERICAL TEST FROM DEFINITION OF DFM

- 1) Compute $\Lambda_{\text{DM}} = \text{Spectrum}(A)$
 - 2) Select randomly $K_y^d \in K^d$
 - 3) Compute $\Lambda_{\text{DFM}} = \Lambda_{\text{DM}} \cap \text{Spectrum}(A + BK_y^d C)$
- ↳ WITH PROBABILITY 1 IT IS THE SET WE REQUIRE!

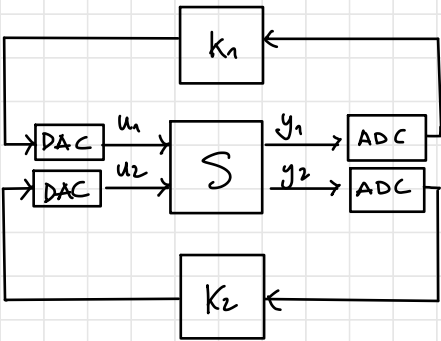
There are two classes of DFMs :

- S-DFM (structural) $\Rightarrow$ cannot be controlled
- NS-DFM (non structural) $\Rightarrow$ can be "controlled" using a decentralized controller

now LINEAR $\downarrow$ LTV $\downarrow$ Time-varying

Infinite-dimensional
(E.G. DELAYS)

DFMs FOR A DIGITAL CONTROLLER



CONTINUOUS-TIME CONTROL:

$$\left. \begin{aligned} u_1(t) &= K_1 y_2(t) \\ u_2(t) &= K_2 y_1(t) \end{aligned} \right\} \begin{aligned} &\exists \text{ FIXED MODES} \\ &\text{(E.G. NON STRUCTURAL)} \end{aligned}$$

DIGITAL CONTROLLER:

$$u_1(t) = K_1 y_2(kh)$$

$$\rightarrow t \in [kh, (k+1)h)$$

$$u_2(t) = K_2 y_1(kh)$$

IT IS NOT AN LTI CONTROLLER
(CONSIDERED FROM THE CONTINUOUS-TIME
SYSTEM "PERSPECTIVE")

$\Rightarrow$ NON-STRUCTURAL FIXED MODES CAN DISAPPEAR!

FOR A DISCRETE-TIME SYSTEM

$$\lambda_{DFH}^{\text{DISCRETE-TIME}} = \bigcap_{K_y^d \in K^d} \text{Spectrum}(F + G K_y^d H)$$

λ_i : MODE IN CONTINUOUS-TIME $\Rightarrow e^{\lambda_i h}$: MODE IN DISCRETE-TIME

IT IS POSSIBLE TO PROVE THAT :

$$\lambda_{CFH}^{\text{DISCRETE-TIME}} \subseteq e^{\lambda_{\text{STRUCT-DFH}} h} \subseteq \lambda_{DFH}^{\text{DISCRETE-TIME} \times h} \subseteq \lambda_{DFH}^{\text{DISCRETE-TIME}} \times h$$

P.S. AS PROVED IN THE REFERENCE

Ü. Özgüner, E.J. Davison, Sampling and Decentralized Fixed Modes,
Proceedings of the 1985 American Control Conference, p.p.257-262

if the system has n distinct eigenvalues, the nonstructural fixed modes are removed for almost all sampling times $h > 0$.

CONTINUOUS TIME

Consider the following system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & -1 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} u_1 + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u_2$$

$$y_1 = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$y_2 = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

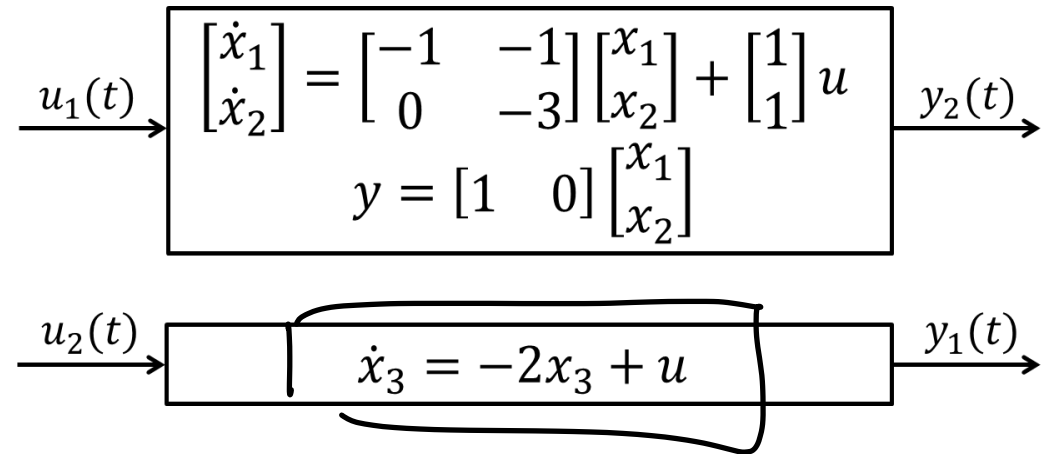
CONTINUOUS TIME

Consider the following system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & -1 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} u_1 + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u_2$$

$$y_1 = [0 \quad 0 \quad 1] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$y_2 = [1 \quad 0 \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$



$$\lambda_3 = -2$$

$$\det \begin{bmatrix} \lambda_3 I - A \\ C_k \end{bmatrix}$$

CANDIDATE SETS : $D = \{1, 2\}$, $A = \{2\}$

$$B_0 = \begin{bmatrix} -1 & +1 & 0 & 1 \\ 0 & +1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \rightarrow \text{rank} = 2 < n$$

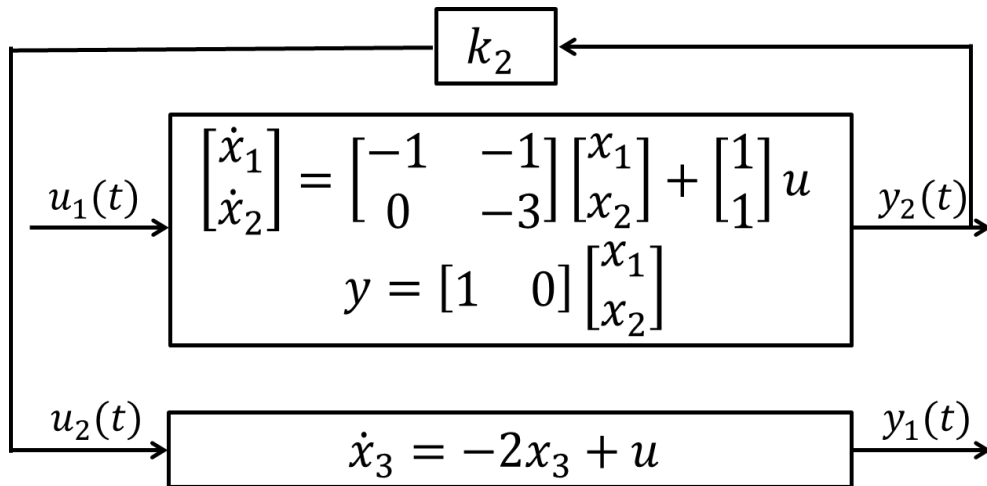
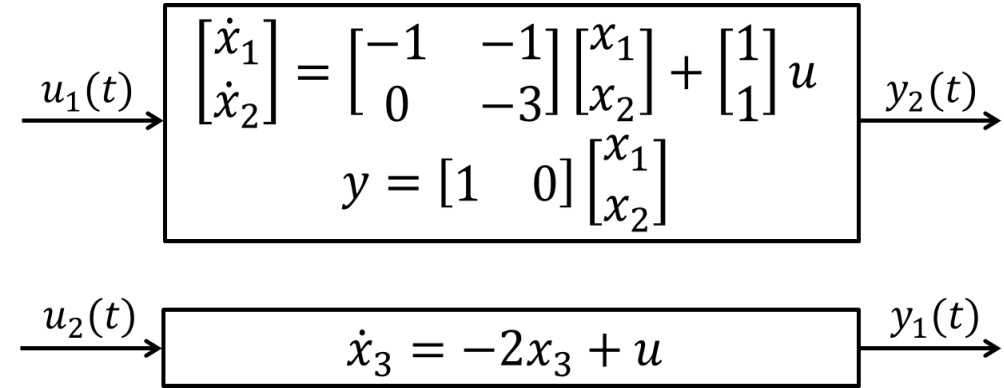
CONTINUOUS TIME

Consider the following system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & -1 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} u_1 + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u_2$$

$$y_1 = [0 \quad 0 \quad 1] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$y_2 = [1 \quad 0 \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$



$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & -1 & 0 \\ 0 & -3 & 0 \\ k_2 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} u_1$$

$$y_1 = [0 \quad 0 \quad 1] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

The reachability matrix of the system above is

$$\mathcal{R} = \begin{bmatrix} 1 & -2 & 5 \\ 1 & -3 & 9 \\ 0 & k_2 & -4k_2 \end{bmatrix}$$

$$\dot{x} = \begin{bmatrix} -1 & -1 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -2 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} u_1 + \begin{bmatrix} 6 \\ 0 \\ 1 \end{bmatrix} K_2 [100] x$$

↓

$$A_{cl}(k_2)$$

$$[\lambda_3 I - A \quad B_1] \rightarrow \text{rank} = 2$$

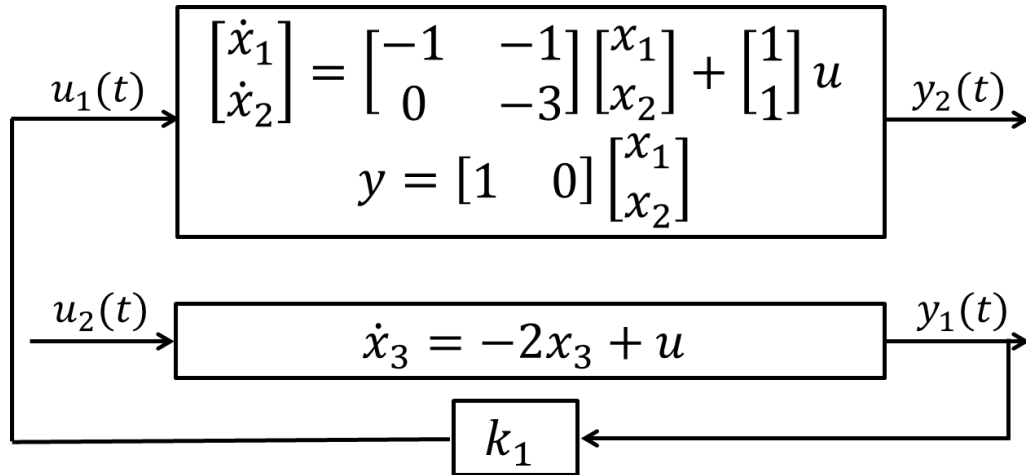
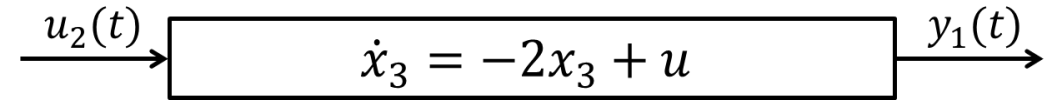
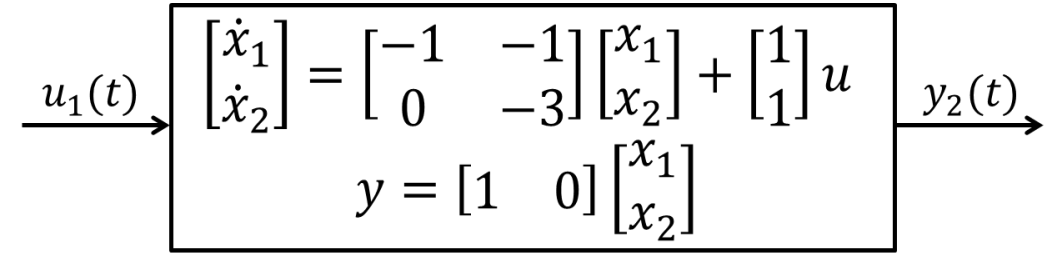
CONTINUOUS TIME

Consider the following system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & -1 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} u_1 + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u_2$$

$$y_1 = [0 \quad 0 \quad 1] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$y_2 = [1 \quad 0 \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$



$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & -1 & k_1 \\ 0 & -3 & k_1 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u_2$$

$$y_2 = [1 \quad 0 \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

The observability matrix of the system above is

$$\mathcal{O} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & -1 & k_1 \\ 1 & 4 & -4k_1 \end{bmatrix}$$

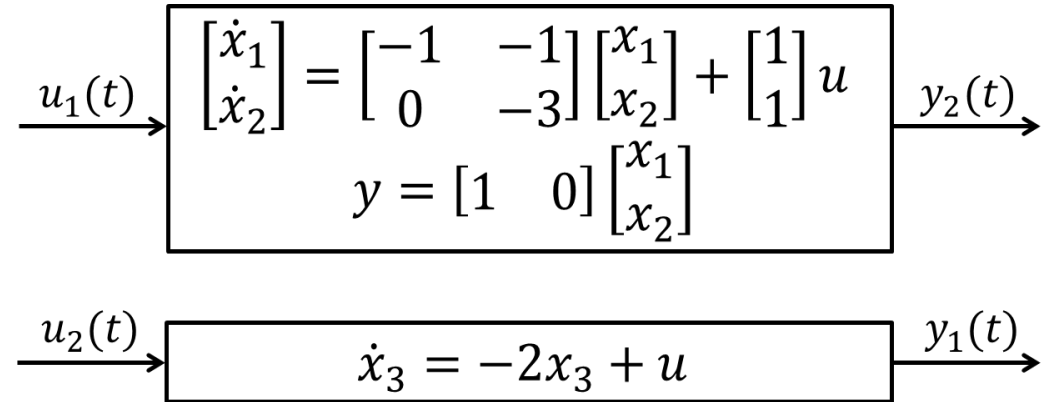
CONTINUOUS TIME

Consider the following system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & -1 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} u_1 + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u_2$$

$$y_1 = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$y_2 = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$



- Consider $\lambda_1 = -1$. The only candidate partition of the index set is $\mathcal{D} = \{2\}$, $\mathcal{H} = \{1\}$. We compute

$$\text{rank} \left(\begin{bmatrix} \lambda_1 I - A & B_{\mathcal{D}} \\ C_{\mathcal{H}} & 0 \end{bmatrix} \right) = \text{rank} \left(\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \right) = 3 = n$$

Therefore, $\lambda_1 = -1$ is not a DFM.

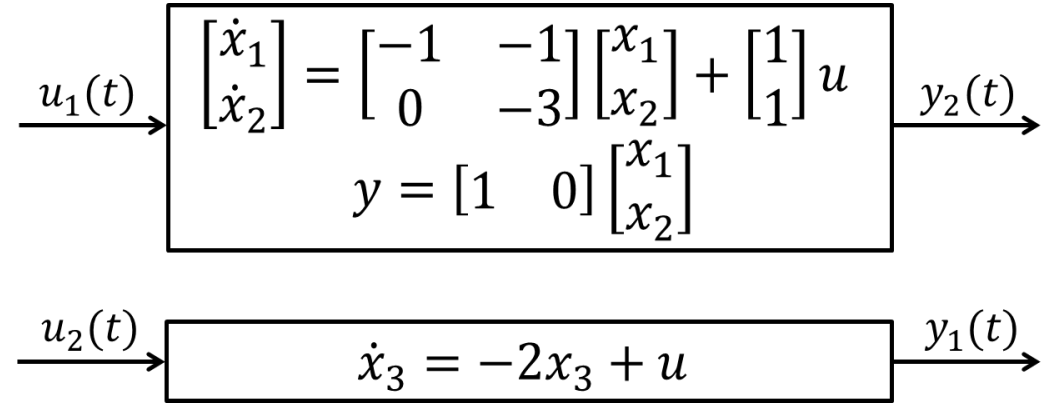
CONTINUOUS TIME

Consider the following system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & -1 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} u_1 + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u_2$$

$$y_1 = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$y_2 = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$



- Consider $\lambda_2 = -3$. The only candidate partition of the index set is $\mathcal{D} = \{2\}$, $\mathcal{H} = \{1\}$. We compute

$$\text{rank} \left(\begin{bmatrix} \lambda_2 I - A & B_{\mathcal{D}} \\ C_{\mathcal{H}} & 0 \end{bmatrix} \right) = \text{rank} \left(\begin{bmatrix} -2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \right) = 3 = n$$

Therefore, $\lambda_2 = -3$ is not a DFM.

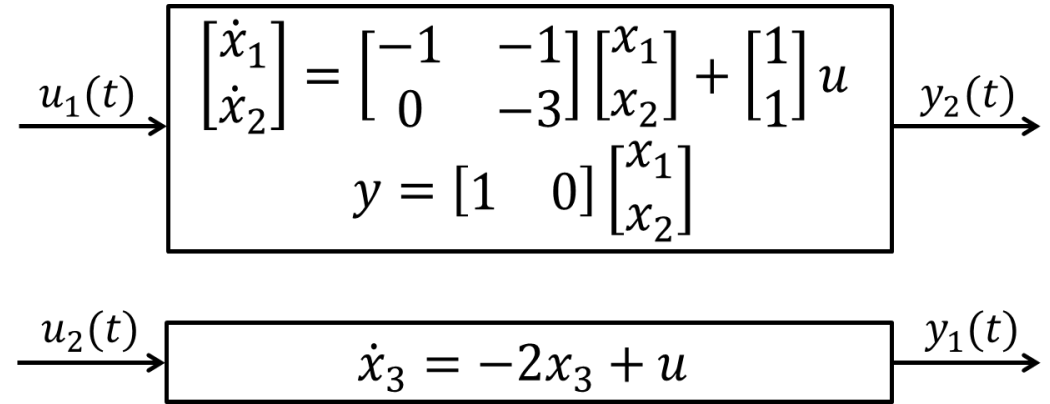
CONTINUOUS TIME

Consider the following system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & -1 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} u_1 + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u_2$$

$$y_1 = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$y_2 = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$



- Consider $\lambda_3 = -2$. The only candidate partition of the index set is $\mathcal{D} = \{1\}$, $\mathcal{H} = \{2\}$. We compute

$$\text{rank} \left(\begin{bmatrix} \lambda_3 I - A & B_{\mathcal{D}} \\ C_{\mathcal{H}} & 0 \end{bmatrix} \right) = \text{rank} \left(\begin{bmatrix} -1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \right) = 2 < n$$

Therefore, $\lambda_3 = -2$ is a DFM.

DISCRETE TIME

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \\ x_3(k+1) \end{bmatrix} = \begin{bmatrix} 0.9048 & -0.0820 & 0 \\ 0 & 0.7408 & 0 \\ 0 & 0 & 0.8187 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix} + \begin{bmatrix} 0.0908 \\ 0.0864 \\ 0 \end{bmatrix} u_1(k) + \begin{bmatrix} 0 \\ 0 \\ 0.0906 \end{bmatrix} u_2(k)$$

$$y_1(k) = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix}$$

$$y_2(k) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix}$$

- Consider $\lambda_1 = 0.9048$. The only candidate partition of the index set is $\mathcal{D} = \{2\}$, $\mathcal{H} = \{1\}$. We compute

$$\text{rank} \left(\begin{bmatrix} \lambda_1 I - F & G_{\mathcal{D}} \\ H_{\mathcal{H}} & 0 \end{bmatrix} \right) = 3 = n$$

Therefore, λ_1 is not a DFM.

DISCRETE TIME

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \\ x_3(k+1) \end{bmatrix} = \begin{bmatrix} 0.9048 & -0.0820 & 0 \\ 0 & 0.7408 & 0 \\ 0 & 0 & 0.8187 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix} + \begin{bmatrix} 0.0908 \\ 0.0864 \\ 0 \end{bmatrix} u_1(k) + \begin{bmatrix} 0 \\ 0 \\ 0.0906 \end{bmatrix} u_2(k)$$

$$y_1(k) = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix}$$

$$y_2(k) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix}$$

- Consider $\lambda_2 = 0.7408$. The only candidate partition of the index set is $\mathcal{D} = \{2\}$, $\mathcal{H} = \{1\}$. We compute

$$\text{rank} \left(\begin{bmatrix} \lambda_2 I - F & G_{\mathcal{D}} \\ H_{\mathcal{H}} & 0 \end{bmatrix} \right) = 3 = n$$

Therefore, λ_2 is not a DFM.

DISCRETE TIME

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \\ x_3(k+1) \end{bmatrix} = \begin{bmatrix} 0.9048 & -0.0820 & 0 \\ 0 & 0.7408 & 0 \\ 0 & 0 & 0.8187 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix} + \begin{bmatrix} 0.0908 \\ 0.0864 \\ 0 \end{bmatrix} u_1(k) + \begin{bmatrix} 0 \\ 0 \\ 0.0906 \end{bmatrix} u_2(k)$$

$$y_1(k) = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix}$$

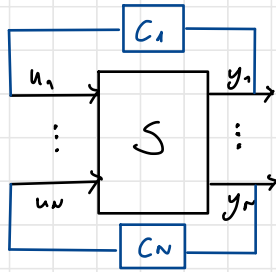
$$y_2(k) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix}$$

- Consider $\lambda_3 = 0.8187$. The only candidate partition of the index set is $\mathcal{D} = \{1\}$, $\mathcal{H} = \{2\}$. We compute

$$\text{rank} \left(\begin{bmatrix} \lambda_3 I - F & G_{\mathcal{D}} \\ H_{\mathcal{H}} & 0 \end{bmatrix} \right) = 3 = n$$

Therefore, λ_3 is not a DFM.

C. DISTRIBUTED CONTROL



u_i IS COMPUTED BASED ON y_i AND ON PIECES OF INFORMATION (OUTPUTS) COMING FROM OTHER CHANNELS

INFORMATION STRUCTURE CONSTRAINTS determine which controller can receive information from which channel

⇒ Neighboring set

$$N_i := \{j \in \mathcal{I} \text{ s.t. } y_j \text{ can be communicated to } C_i\}$$

↓
FOR COMPUTING u_i

We also define

$$\mathcal{N} := \{N_1, \dots, N_N\} \quad \text{"INFORMATION STRUCTURE CONSTRAINTS SET"}$$

1) STATIC OUTPUT FEEDBACK CONTROL LAW

$$u_i = \sum_{j \in N_i} K_y^{ij} y_j$$

$$= \begin{bmatrix} K_y^{i1} & K_y^{i2} & \dots & K_y^{iN} \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix}$$

$$K_y^{ij} = 0 \quad \text{if } j \notin N_i$$

$$u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_N \end{bmatrix} = \begin{bmatrix} K_y^{11} & K_y^{12} & \dots & K_y^{1N} \\ K_y^{21} & K_y^{22} & \dots & K_y^{2N} \\ \vdots & \vdots & \ddots & \vdots \\ K_y^{N1} & K_y^{N2} & \dots & K_y^{NN} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix} \stackrel{(*)}{=} K_y y$$

- IN THE CENTRALIZED CASE: $K_y \in \mathbb{R}^{u \times p}$
- IN THE DECENTRALIZED CASE: $K_y \in \mathcal{K}^y$ (set of suitable block-diagonal matrices)
- IN THE DISTRIBUTED CASE: K_y HAS A "SPARSE" STRUCTURE, COMPATIBLE WITH THE INFORMATION STRUCTURE CONSTRAINTS

$\Downarrow$
 $K_y \in \mathcal{K}^{N^y}$ (set of sparse matrices consistent with the information structure constraints)

CLOSED-LOOP DYNAMICS:

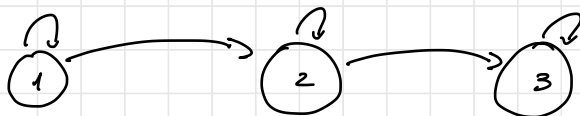
$$\dot{x} = Ax + Bu = (A + BK_y C)x$$

$\hookrightarrow K_y \in \mathcal{K}^N$

EXAMPLES

WE USE A GRAPH-BASED REPRESENTATION

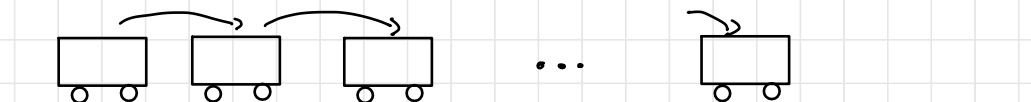
I) STRING TOPOLOGY
 (UNIDIRECTIONAL)



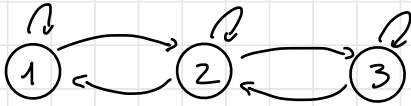
$$\begin{aligned} u_1 &= K_y^{11} y_1 \\ u_2 &= K_y^{22} y_2 + K_y^{21} y_1 \\ u_3 &= K_y^{33} y_3 + K_y^{32} y_2 \end{aligned}$$

$$\Rightarrow \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} K_y^{11} & 0 & 0 \\ K_y^{21} & K_y^{22} & 0 \\ 0 & K_y^{32} & K_y^{33} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

MATRIX:
 • BANDED
 • TRIANGULAR



II) BIDIRECTIONAL STRING

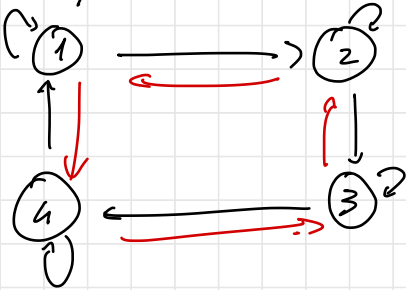


$$\begin{cases} u_1 = K_y^{11} y_1 + K_y^{12} y_2 \\ u_2 = K_y^{21} y_1 + K_y^{22} y_2 + K_y^{23} y_3 \\ u_3 = K_y^{33} y_3 + K_y^{32} y_2 \end{cases} \Rightarrow K_y = \begin{bmatrix} K_y^{11} & K_y^{12} & 0 \\ K_y^{21} & K_y^{22} & K_y^{23} \\ 0 & K_y^{32} & K_y^{33} \end{bmatrix}$$

• BANDED

• NOT TRIANGULAR

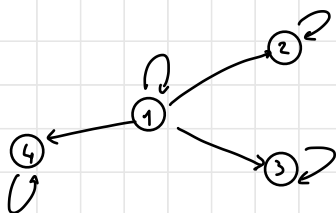
III) CYCLE



$$\begin{cases} u_1 = K_y^{11} y_1 + K_y^{14} y_4 + K_y^{12} y_2 \\ u_2 = K_y^{22} y_2 + K_y^{21} y_1 + K_y^{23} y_3 \\ u_3 = K_y^{33} y_3 + K_y^{32} y_2 + K_y^{34} y_4 \\ u_4 = K_y^{44} y_4 + K_y^{43} y_3 + K_y^{41} y_1 \end{cases}$$

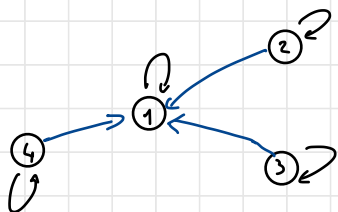
$$\Rightarrow K_y = \begin{bmatrix} K_y^{11} & \textcircled{K_y^{12}} & 0 & K_y^{14} \\ K_y^{21} & K_y^{22} & \textcircled{K_y^{23}} & 0 \\ 0 & K_y^{32} & K_y^{33} & \textcircled{K_y^{34}} \\ \textcircled{K_y^{41}} & 0 & K_y^{43} & K_y^{44} \end{bmatrix}$$

IV) STAR



$$\begin{cases} u_1 = K_y^{11} y_1 \\ u_2 = K_y^{22} y_2 + K_y^{21} y_1 \\ u_3 = K_y^{33} y_3 + K_y^{31} y_1 \\ u_4 = K_y^{44} y_4 + K_y^{41} y_1 \end{cases}$$

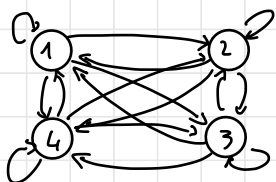
$$K_y = \begin{bmatrix} K_y^{11} & 0 & 0 & 0 \\ K_y^{21} & K_y^{22} & 0 & 0 \\ K_y^{31} & 0 & K_y^{33} & 0 \\ K_y^{41} & 0 & 0 & K_y^{44} \end{bmatrix} \rightsquigarrow \text{Bordered block diagonal}$$



$$\begin{cases} u_1 = K_y^{11} y_1 + K_y^{12} y_2 + K_y^{13} y_3 + K_y^{14} y_4 \\ u_2 = K_y^{22} y_2 \\ u_3 = K_y^{33} y_3 \\ u_4 = K_y^{44} y_4 \end{cases}$$

$$K_y = \begin{bmatrix} K_y^{11} & K_y^{12} & K_y^{13} & K_y^{14} \\ 0 & K_y^{22} & 0 & 0 \\ 0 & 0 & K_y^{33} & 0 \\ 0 & 0 & 0 & K_y^{44} \end{bmatrix} \rightsquigarrow \text{Bordered block diagonal}$$

V) ALL-TO-ALL



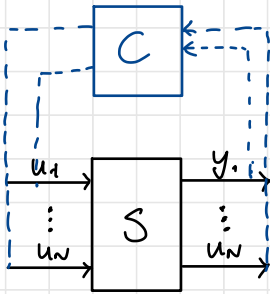
$$\begin{cases} u_1 = K_y^{11} y_1 + K_y^{12} y_2 + K_y^{13} y_3 + K_y^{14} y_4 \\ u_2 = K_y^{21} y_1 + K_y^{22} y_2 + K_y^{23} y_3 + K_y^{24} y_4 \\ u_3 = K_y^{31} y_1 + K_y^{32} y_2 + K_y^{33} y_3 + K_y^{34} y_4 \\ u_4 = K_y^{41} y_1 + K_y^{42} y_2 + K_y^{43} y_3 + K_y^{44} y_4 \end{cases}$$

$$K_y = \begin{bmatrix} K_y^{11} & K_y^{12} & K_y^{13} & K_y^{14} \\ K_y^{21} & K_y^{22} & K_y^{23} & K_y^{24} \\ K_y^{31} & K_y^{32} & K_y^{33} & K_y^{34} \\ K_y^{41} & K_y^{42} & K_y^{43} & K_y^{44} \end{bmatrix}$$

CENTRALIZED CONTROL

OPTIMAL PERFORMANCE

1 CONTROLLER



STAR-LIKE COMMUNICATION,
WHERE C LIES IN THE
CENTER

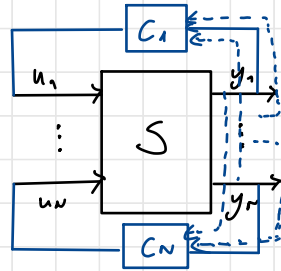


NOT FLEXIBLE
NOT RESILIENT

DISTRIBUTED ALL-TO-ALL CONTROL

OPTIMAL PERFORMANCE

N CONTROLLERS



ALL-TO-ALL
COMMUNICATION



FLEXIBLE
POSSIBLY RESILIENT

2) DYNAMIC OUTPUT-FEEDBACK CONTROL LAW

$C_1, \dots, C_N$ ARE DYNAMICAL SYSTEMS

$$\begin{array}{c} y_1 \\ \vdots \\ y_N \end{array} \rightarrow \boxed{C_i} \xrightarrow{u_i} \Rightarrow C_i: \begin{cases} \dot{\bar{x}}_i = A_c^i \bar{x}_i + \sum_{j \in N_i} B_c^{ij} y_j \\ u_i = K_x^i \bar{x}_i + \sum_{j \in N_i} K_y^{ij} y_j \end{cases}$$

$\downarrow$
 y_j s.t. $j \in N_i$

$$\bar{x} = \begin{bmatrix} \bar{x}_1 \\ \vdots \\ \bar{x}_N \end{bmatrix} \quad u = \begin{bmatrix} u_1 \\ \vdots \\ u_N \end{bmatrix} \quad y = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix}$$

$$\begin{cases} \dot{\bar{x}} = \begin{bmatrix} \dot{\bar{x}}_1 \\ \vdots \\ \dot{\bar{x}}_N \end{bmatrix} = \underbrace{\begin{bmatrix} A_c^1 & 0 & \dots & 0 \\ 0 & A_c^2 & 0 & \dots \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & A_c^N \end{bmatrix}}_{\text{BLOCK-DIAGONAL}} \bar{x} + \begin{bmatrix} B_c^{11} & B_c^{12} & \dots & B_c^{1N} \\ B_c^{21} & B_c^{22} & \dots & B_c^{2N} \\ \vdots & \vdots & \ddots & \vdots \\ B_c^{N1} & B_c^{N2} & \dots & B_c^{NN} \end{bmatrix} y \\ u = \begin{bmatrix} K_x^1 & 0 & \dots & 0 \\ 0 & K_x^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & K_x^N \end{bmatrix} \bar{x} + \begin{bmatrix} K_y^{11} & K_y^{12} & \dots & K_y^{1N} \\ \vdots & \vdots & \ddots & \vdots \\ K_y^{N1} & K_y^{N2} & \dots & K_y^{NN} \end{bmatrix} y \end{cases} \rightarrow L \in K^N$$

3) STATIC STATE-FEEDBACK CONTROL LAW

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_N \end{bmatrix} \rightarrow \text{non-overlapping subvectors}$$

$$\begin{aligned} u_i &= \sum_{j \in N_i} K_x^{ij} x_j = [K_x^{i1} \ K_x^{i2} \ \dots \ K_x^{iN}] x \\ &= K_x x, \quad K_x \in K^N \Rightarrow \dot{x} = (A + BK_x) x \end{aligned}$$

$\rightarrow K_x^{ij} = 0$ IF $j \notin N_i$

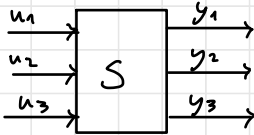
As long as some blocks of K_x are $= 0$
 $\Rightarrow$ This limits the possibility of controlling the system properly

$\Rightarrow$ Not possible to assign the closed-loop system eigenvalues at will

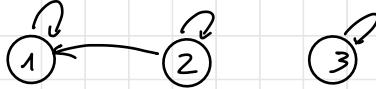
4) DISTRIBUTED FIXED MODES (DFM)

In the literature, distributed fixed modes are also called "decentralized overlapping fixed modes". This is due to the fact that a distributed control structure can be regarded as a decentralized control structure with overlapping outputs.

FOR EXAMPLE: CONSIDER A SYSTEM WITH 3 CHANNELS



AND THE DISTRIBUTED CONTROL TOPOLOGY:



The control law is:

$$\begin{cases} u_1 = K_1^{11} y_1 + K_1^{12} y_2 \\ u_2 = K_2^{22} y_2 \\ u_3 = K_3^{33} y_3 \end{cases}$$

This is equivalent to a decentralized controller applied to a system where the first channel has output $y_1 = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$, which overlaps with the output of channel 2. In fact, in this case:

The decentralized controller is

$$\begin{cases} u_1 = K_1^1 y_1 = [K_1^{11} \ K_1^{12}] \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \\ u_2 = K_2^2 y_2 \\ u_3 = K_3^3 y_3 \end{cases}$$

Questions:

- IS THERE A CONDITION THAT ALLOWS TO PLACE THE CLOSED-LOOP SYSTEM EIGENVALUES IN DESIRED POSITIONS?
- IN CASE THE FORMER CONDITION IS SATISFIED, WHICH TYPE OF CONTROL LAW ALLOWS US TO PLACE THE EIGENVALUES?

Definition: The set

$$\Lambda_{\text{DIFM}} = \bigcap_{K_y \in \mathcal{K}^n} \text{spectrum}(A + BK_y C)$$

is the set of distributed fixed modes

Theorem:

- IF THE SYSTEM HAS NO **DISTRIBUTED** FIXED MODES
 $\Rightarrow$ ALL EIGENVALUES CAN BE PLACED IN ARBITRARY DESIRED POSITIONS USING A **DISTRIBUTED** (LINEAR, TIME-INVARIANT, FINITE-DIMENSIONAL) **DYNAMIC OUTPUT FEEDBACK CONTROL LAW**
- WITH A LINEAR, TIME-INVARIANT, FINITE-DIMENSIONAL DECENTRALIZED CONTROLLER WE CANNOT MOVE **DIFMs**

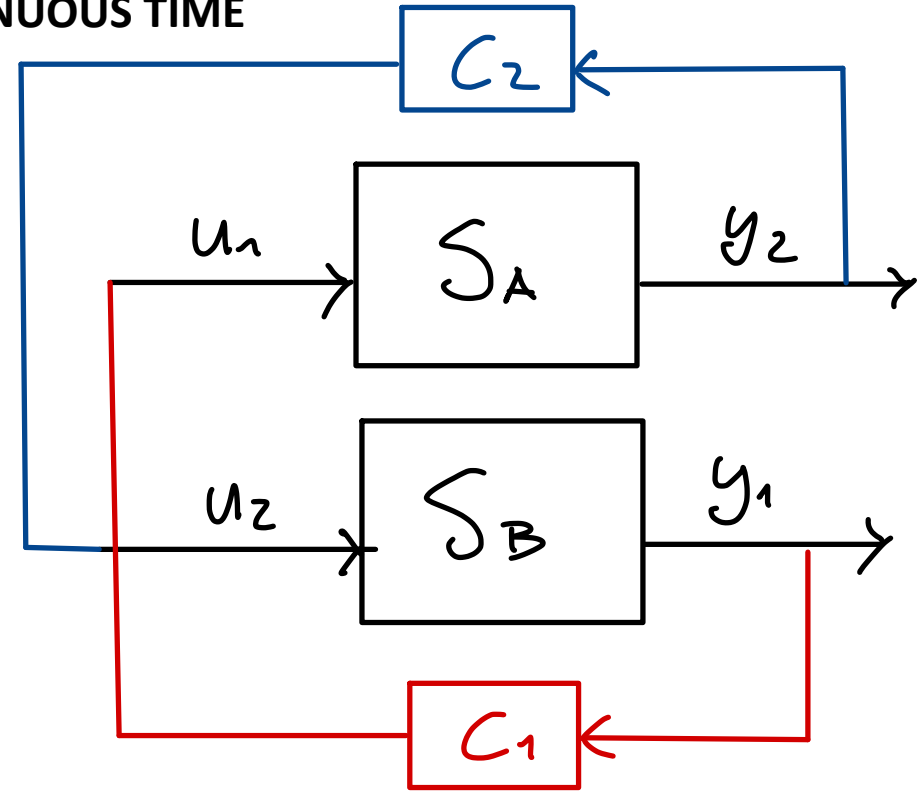
DISTRIBUTED - CONTINUOUS TIME

Consider the following system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & -1 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} u_1 + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u_2$$

$$y_1 = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$y_2 = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$



1. We define $\mathcal{N}_1 = \{1,2\}$ and $\mathcal{N}_2 = \{2\}$. Indeed, we allow matrix K_y to be of the form

$$K_y = \begin{bmatrix} K_y^{11} & K_y^{12} \\ 0 & K_y^{22} \end{bmatrix}$$

In this case $\lambda_3 = -2$ is a DiFM.

DISTRIBUTED - CONTINUOUS TIME

Consider the following system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & -1 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} u_1 + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u_2$$

$$y_1 = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$y_2 = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

2. We define $\mathcal{N}_2 = \{1\}$ and $\mathcal{N}_2 = \{1,2\}$, allowing matrix K_y to be of the form

$$K_y = \begin{bmatrix} K_y^{11} & 0 \\ K_y^{21} & K_y^{22} \end{bmatrix}$$

In this case the system has no DiFMs.

3. DESIGN OF STATIC STATE-FEEDBACK CONTROL

LAWS USING LMIS

A) LMIS FOR STABILITY (CENTRALIZED CASE)

1) CONTINUOUS-TIME SYSTEMS

$$\dot{x} = Ax + Bu \quad u = Kx$$

$$\dot{x} = (A + BK)x$$

$\Rightarrow \exists Kx$ stabilizing if (A, B) is controllable

FROM THE LYAPUNOV THEOREM:

$A_{cl} = A + BKx$ is Hurwitz stable iff $\exists Kx, P = P^T > 0$ such that

$$(A + BKx)^T P + P(A + BKx) < 0 \quad \text{We define } y = P^{-1}$$

$$\downarrow$$
$$y(A^T P + P A + \underbrace{Kx^T B^T P + P B Kx}_{\text{NOT LINEAR!}}) y < 0$$

$$\downarrow$$
$$y A^T + A y + \underbrace{y Kx^T B}_{L^T} + \underbrace{B Kx y}_{L} < 0$$

(HAS THE SAME DIMENSION WITH RESPECT TO Kx , I.E. THE SAME NUMBER OF DEGREES OF FREEDOM)

$$\Rightarrow y A^T + A y + L^T B + B L < 0 \quad \text{LMI}$$

THE CORRESPONDING LMI FEASIBILITY PROBLEM IS:

FIND L, Y S.T.

$$YA^T + AY + L^T B + BL < 0$$

IF FEASIBLE $\Rightarrow Y, L$ (Results) $\Rightarrow K_x = LY^{-1}$

$\Rightarrow A+BK_x$ IS HURWITZ STABLE

2) CONTINUOUS-TIME SYSTEMS

$$\begin{aligned} x_{k+1} &= Fx_k + Gu_k, \quad u_k = Kx_k \\ &= (F + GK_x)x_k \end{aligned}$$

FROM THE LYAPUNOV THEOREM:

$F_{cl} = F + GK_x$ IS SCHUR STABLE IFF $\exists K_x, P=P^T > 0$ S.T.

$$(F + GK_x)P(F + GK_x)^T - P < 0$$

$\downarrow$

$$FPF^T - P + \underbrace{GK_x P F}_{\downarrow L} + \underbrace{F P K_x^T}_{\downarrow L^T} G^T + \underbrace{GK_x P K_x^T G^T}_{\substack{\downarrow G K_x P P^{-1} P K_x^T G^T \\ \downarrow \underbrace{G K_x P P^{-1}}_{\downarrow L} \underbrace{P K_x^T G^T}_{\downarrow L^T}}} < 0$$

$$\Rightarrow FPF^T - P + GLF + FL^T G + GLP^{-1}L^T G^T < 0$$

$$\underbrace{P - FPF^T - GLF - FL^T G}_Q - \underbrace{GL}_{S} \underbrace{P^{-1}}_{R^{-1}} \underbrace{L^T G^T}_{S^T} > 0 \quad (*)$$

We use the Schur complement

$$\begin{cases} Q - SR^{-1}S^T > 0 \\ R > 0 \end{cases} \Leftrightarrow \begin{bmatrix} Q & S \\ S^T & R \end{bmatrix} > 0$$

Thanks to the Schur complement, (*) can be rewritten as:

$$\begin{bmatrix} P - FPF^T - FL^T G^T - GLF & GL \\ L^T G^T & P \end{bmatrix} > 0 \quad (i)$$

THE CORRESPONDING LMI FEASIBILITY PROBLEM IS:

FIND L, P S.T. (i) IS VERIFIED

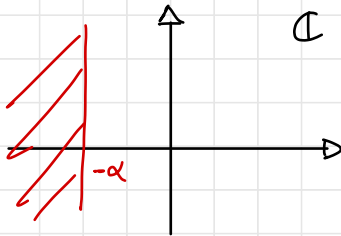
IF FEASIBLE $\Rightarrow P, L$ (Results) $\Rightarrow K_x = LP^{-1}$

$\Rightarrow A+BK_x$ IS HURWITZ STABLE

B) LMIS FOR PERFORMANCE (CENTRALIZED CASE)

1) CONDITION ON THE SPECTRAL ABSCISSA

GIVEN A CONTINUOUS-TIME SYSTEM (A, B) AND THE CONTROL LAW $u = Kx$



IN CLOSED LOOP

$$\begin{aligned}\dot{x} &= A x + B K x \\ &= (A + B K) x\end{aligned}$$

We want to force the eigenvalues of $A + B K$ to be such that $\text{Re}(\lambda) < -\alpha$

This is enforced if $A + B K + \alpha I$ is Hurwitz stable. In fact

$$\text{Re}(\text{eig}(A + B K + \alpha I)) = \text{Re}(\text{eig}(A + B K)) + \alpha < 0$$

$\Downarrow$

$$\text{Re}(\text{eig}(A + B K)) < -\alpha$$

This is done using the Lyapunov inequality applied to $A + B K + \alpha I$

$$\Rightarrow Y(A + B K + \alpha I)^T + (A + B K + \alpha I)Y < 0$$

$\Updownarrow$

$$Y(A + B K)^T + (A + B K)Y + 2\alpha Y < 0$$

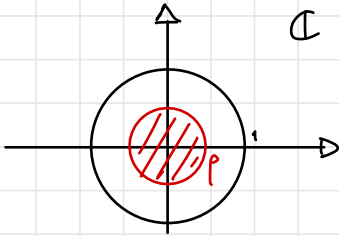
$\Updownarrow$

$$YA^T + AY + L^T B^T + BL + 2\alpha Y < 0$$

LMi

2) CONDITION ON THE SPECTRAL RADIUS

GIVEN A DISCRETE-TIME SYSTEM (F, G) AND THE CONTROL LAW $u = Kx$ IN CLOSED LOOP



$$\begin{aligned} x_{k+1} &= Fx_k + G K x_k \\ &= (F + G K) x_k \end{aligned}$$

We want to force the eigenvalues of $F + G K$ to be such that $|\lambda| < \rho$

This is enforced if $\frac{1}{\rho}(F + G K)$ is Schur stable.

This is done using the Lyapunov inequality applied to $\frac{1}{\rho}(F + G K)$:

$$\rho^2 \left(\frac{1}{\rho^2} (F + G K) P (F + G K)^T - P \right) < 0$$



$$(F + G K) P (F + G K)^T - \rho^2 P < 0$$



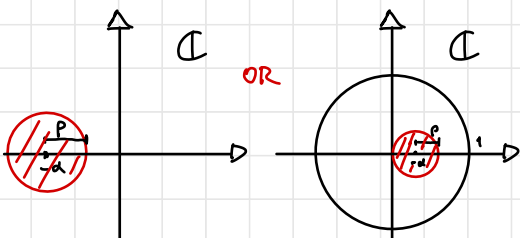
$$\begin{bmatrix} \rho^2 P - F P F^T - F L^T G^T - G L F & G L \\ L^T G^T & P \end{bmatrix} > 0$$

ONLY DIFFERENT WITH
RESPECT TO THE
CASE OF SCHUR
STABILITY

Where $L = K P$

3) Eigenvalues in a circle with radius ρ and center in $-\alpha$
(USABLE BOTH IN CONTINUOUS-TIME AND DISCRETE-TIME)

GIVEN A SYSTEM (A, B) AND THE CONTROL LAW $u = Kx$,



IN CLOSED LOOP

$$\begin{aligned}\dot{x} &= Ax + B K x \\ &= (A + B K) x\end{aligned}$$

We want to force the eigenvalues of $A + B K$ to be in a circle with radius ρ and center in $-\alpha$

This is enforced if $\frac{1}{\rho}(F + GKx + \alpha I)$ is Schur stable.

This is done using the Lyapunov inequality applied to $\frac{1}{\rho}(F + GKx + \alpha I)$:

$$(A + B K x + \alpha I) P (A + B K x + \alpha I)^T - \rho^2 P < 0$$

$$(A + B K x) P (A + B K x)^T + \alpha [P(A + B K x)^T + (A + B K x) P] + \alpha^2 P - \rho^2 P < 0$$

$-(\rho^2 - \alpha^2)P$

We set $L = K P$

$$(\rho^2 - \alpha^2)P - A P A^T - B L A^T - A^T L^T B^T - \alpha [P A^T + L^T B^T + A P + B L] - B L P^T L^T B^T > 0$$

Using the Schur complement we obtain:

$$\begin{bmatrix} (\rho^2 - \alpha^2)P - A P A^T - B L A^T - A^T L^T B^T - \alpha [P A^T + L^T B^T + A P + B L] & B L \\ L^T B^T & P \end{bmatrix} > 0$$

4) Eigenvalues in a region D (D-stability)

GIVEN A SYSTEM (A, B) AND THE CONTROL LAW $u = Kx$,
IN CLOSED LOOP

$$\dot{x} = Ax + BKx = (A + BK)x$$

We want to force the eigenvalues of $A + BK$ to verify

$$\lambda \in D := \{z \in \mathbb{C} : f_D(z) < 0\} \text{ where}$$

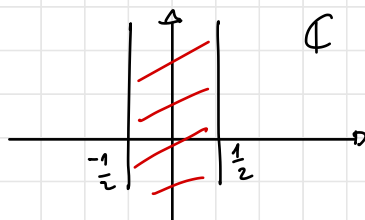
$$f_D(z) = \underbrace{A}_{\text{symmetric, square, real}} + \underbrace{\Theta}_{\text{square, real}} z + \underbrace{\Theta^T}_{\text{complex conjugate}} z^*$$

EXAMPLES

a) $A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$, $\Theta = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ ($z, z^* = z \operatorname{Re}(z)$)

$$f_D(z) = \begin{bmatrix} -1 + z + z^* & 0 \\ 0 & -1 - z - z^* \end{bmatrix} = \begin{bmatrix} -1 + 2\operatorname{Re}(z) & 0 \\ 0 & -1 - 2\operatorname{Re}(z) \end{bmatrix} < 0$$

$$\Rightarrow \begin{cases} -1 + 2\operatorname{Re}(z) < 0 \\ -1 - 2\operatorname{Re}(z) < 0 \end{cases} \Rightarrow -\frac{1}{2} < \operatorname{Re}(z) < \frac{1}{2}$$



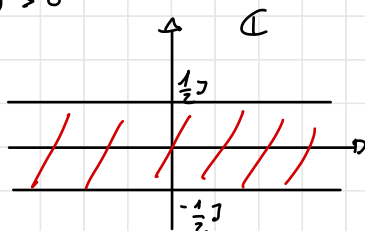
b) $A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$, $\Theta = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

$$f_D(z) = A + \Theta z + \Theta^T z^* = \begin{bmatrix} -1 & z - z^* \\ -z + z^* & -1 \end{bmatrix} = \begin{bmatrix} -1 & 2j\operatorname{Im}(z) \\ -2j\operatorname{Im}(z) & -1 \end{bmatrix} < 0$$

($z \cdot z^* = z j \operatorname{Im}(z)$)

$$\begin{cases} \operatorname{tr}(f_D(z)) < 0 & -1 - 1 < 0 \quad ? \quad \text{yes} \\ \det(f_D(z)) > 0 & \det(f_D(z)) = 1 - 4\operatorname{Im}^2(z) > 0 \end{cases}$$

$$\Rightarrow |\operatorname{Im}(z)| < \frac{1}{4}$$



$$c) \quad \mathcal{L} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \Theta = \begin{bmatrix} \sin(\theta) & \cos(\theta) \\ -\cos(\theta) & \sin(\theta) \end{bmatrix}$$

$$\Rightarrow f_0(z) = \begin{bmatrix} \sin(\theta)(z+z^*) & \cos(\theta)(z-z^*) \\ \cos(\theta)(z^*-z) & \sin(\theta)(z+z^*) \end{bmatrix}$$

$$= \begin{bmatrix} 2\sin(\theta)\operatorname{Re}(z) & 2j\cos(\theta)\operatorname{Im}(z) \\ -2j\cos(\theta)\operatorname{Im}(z) & 2\sin(\theta)\operatorname{Re}(z) \end{bmatrix} < 0$$

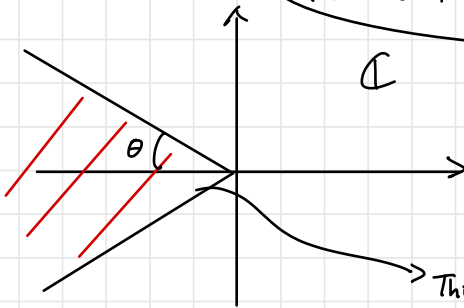
$$\begin{cases} \operatorname{tr}(f_0(z)) = 4 \underbrace{\sin(\theta)}_{>0} \operatorname{Re}(z) < 0 & \theta \in \left(\frac{\pi}{2}, \pi \right) \\ \det(f_0(z)) = 4\sin^2(\theta)\operatorname{Re}^2(z) - 4\cos^2(\theta)\operatorname{Im}^2(z) > 0 \end{cases}$$

$$\Downarrow$$

$$\operatorname{Re}(z) < 0 \quad (i)$$

$$\Downarrow$$

$$|\operatorname{Im}(z)| < |\tan(\theta)| |\operatorname{Re}(z)| \quad (ii)$$



IN CONTINUOUS-TIME $\Rightarrow$ MODES $e^{\lambda t}$

$$\lambda = \alpha + j\omega = \zeta\omega_n + j\sqrt{1-\zeta^2}\omega_n$$

where $\zeta \in [0, 1)$

This implies that $\zeta \in (\cos(\theta), 1]$!

REMARK: BY COMBINING THE REQUIREMENT CONSIDERED IN POINT 1) [i.e. $\operatorname{Re}(\lambda) < -\alpha$] AND THE ONE CONSIDERED HERE [i.e. $\operatorname{Re}(\lambda) < \tan(\alpha)\operatorname{Im}(\lambda)$] WE CAN JOINTLY IMPOSE THE SETTLING TIME AND THE MAXIMUM OVERSHOOT TO THE SYSTEM (STEP) RESPONSE. IN FACT

$$S\% = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$$

$$T_{\text{settling}} \approx \frac{5}{\omega_n\zeta}$$

Theorem $\lambda_i \in \mathbb{D}$ [for all $i=1, \dots, n$] IF $\exists \gamma = \gamma^T > 0$ SUCH THAT

$$M_D < 0$$

Matrix M_D is defined as follows:

- recall that, in closed loop ($u = Kx$), $\dot{x} = (A + BK)x$
- each block (i,j) of M_D has the following definition,

$$M_D|_{ij} = \lambda_{ij} + \theta_{ij} (A + BK)^T \gamma + \theta_{ji} \gamma (A + BK)^T$$

λ_{ij} : (i,j) -th element of Λ

θ_{ij} : (i,j) -th element of Θ

CONSIDER EXAMPLE C)

$$\Lambda = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \Theta = \begin{bmatrix} \sin(\theta) & \cos(\theta) \\ -\cos(\theta) & \sin(\theta) \end{bmatrix}$$

$$M_D = \begin{bmatrix} \sin(\theta)(A+BK)^T \gamma + \sin(\theta) \gamma (A+BK)^T & \cos(\theta)(A+BK)^T \gamma - \cos(\theta) \gamma (A+BK)^T \\ -\cos(\theta)(A+BK)^T \gamma + \cos(\theta) \gamma (A+BK)^T & \sin(\theta)(A+BK)^T \gamma + \sin(\theta) \gamma (A+BK)^T \end{bmatrix}$$

$$= \begin{bmatrix} \sin(\theta) \left[(A+BK)^T \gamma + \gamma (A+BK)^T \right] & \cos(\theta) \left[(A+BK)^T \gamma - \gamma (A+BK)^T \right] \\ -\cos(\theta) \left[(A+BK)^T \gamma - \gamma (A+BK)^T \right] & \sin(\theta) \left[(A+BK)^T \gamma + \gamma (A+BK)^T \right] \end{bmatrix} < 0$$

IT IS AN LMI IF WE DEFINE $L = K^T \gamma$, I.E.,

$$M_D = \begin{bmatrix} \sin(\theta) \left[AY + YA^T + BL + L^T B^T \right] & \cos(\theta) \left[AY - YA^T + BL - L^T B^T \right] \\ -\cos(\theta) \left[AY - YA^T + BL - L^T B^T \right] & \sin(\theta) \left[AY + YA^T + BL + L^T B^T \right] \end{bmatrix} < 0$$

5) COMPUTATIONAL BURDEN

IN THE PREVIOUS LMI PROBLEMS THE COMPUTATIONAL EFFORT IS PROPORTIONAL TO THE NUMBER OF FREE VARIABLES

FREE VARIABLES: γ (OR P) $\rightarrow$ SQUARE, SYMMETRIC MATRIX, $n \times n$

$$n + (n-1) + (n-2) + \dots + 1 = \frac{(n+1)n}{2} \quad \text{FREE VARIABLES}$$

$\bullet L \in \mathbb{R}^{m \times n} \rightarrow m \times n$ free variables

$$\Rightarrow \eta_{\text{centr.}} = nm + n \frac{(n+1)}{2}$$

very large for large-scale systems !!

6) Reduction of the control effort

Besides stability, we may want to limit the gain of the controller i.e., for a given K_x

$$\|K_x\| \leq K_x \quad \underline{K_x = LY^{-1}} \quad (\text{OR } K_x = LP^{-1})$$

$$\leq \|LY^{-1}\| \leq \|L\| \|Y^{-1}\| \leq K_x$$

CONSIDER $\|L\|$ AND $\|Y^{-1}\|$ SEPARATELY.

$$\bullet \|L\| \leq \sqrt{K_L} \Leftrightarrow L^T L \leq K_L I$$



$$K_L I - L^T L \geq 0$$

$\Leftrightarrow$

$$\begin{bmatrix} K_L I & L^T \\ L & I \end{bmatrix} \begin{matrix} C_1 \\ C_2 \end{matrix} \geq 0$$

[SEE THE SQUIR

COMPLEMENT BELOW]

$$Q = K_L I, \quad S = L^T, \quad R^{-1} = I$$

SCHUR COMPLEMENT

$$\begin{cases} Q - S R^{-1} S^T \geq 0 \\ R > 0 \end{cases} \Leftrightarrow \begin{bmatrix} Q & S \\ S^T & R \end{bmatrix} \geq 0$$

$$\|Y^{-1}\| \leq K_L \quad Y^{-1} \leq K_Y I \quad \Leftrightarrow \begin{bmatrix} K_Y I & I \\ I & Y \end{bmatrix} \geq 0 \quad (ii')$$

$$K_Y I - Y^{-1} \geq 0$$

[SEE THE SCHUR COMPLEMENT ABOVE] $K_Y I = Q \quad S = S^T = I \quad R = Y$

LM1 OPTIMIZATION

PROBLEM: $\min_{L, Y, K_Y, K_L} J = \alpha_Y K_Y + \alpha_L K_L \quad \text{s.t.} \quad \begin{aligned} &\bullet (i), (ii') \\ &\bullet \text{inequalities for stability} \end{aligned}$

7) Minimization of the H_2 norm [continuous-time]

CONSIDER THE SYSTEM WITH ADDITIVE NOISE W

$$\begin{cases} \dot{x} = Ax + Bu + B_w W = (A + BK_x)x + B_w W \\ z = Cx + Du = (C + DK_x)x \end{cases}$$

PROBLEM: find K_x such that $u = K_x x$ allows to minimize the H_2 norm of the transfer function $G_{zw}(s)$ from w to z

Theorem

$$\|G_{zw}(s)\|_2^2 = \inf \left\{ \text{tr} \left\{ (C + DK_x) Y (C + DK_x)^T \right\} \right.$$

$$\left. \text{s.t. } (A + BK_x) Y + Y (A + BK_x)^T + B_w B_w^T < 0 \right\}$$

$$L = K_x Y$$

$$\min \text{tr} (C + DK_x) Y (C + DK_x)^T = (C + DK_x) Y Y^T Y (C + DK_x)^T$$

$$\downarrow$$

$$\boxed{\min \text{tr}(S)}$$

$$\begin{aligned} &= (Cy + DK_x Y) Y^{-1} (Cy + DK_x Y)^T \\ &= (Cy + DL) Y^{-1} (Cy + DL)^T \end{aligned}$$

$$S - (Cy + DL) Y^{-1} (Cy + DL)^T \geq 0$$

$$\downarrow$$

$$\boxed{\begin{bmatrix} S & Cy + DL \\ (Cy + DL)^T & Y \end{bmatrix} \geq 0}$$

$$\text{LMI: } (A + BK_x) Y + Y (A + BK_x)^T + B_w B_w^T < 0$$

$$\downarrow$$

$$\boxed{AY + YA^T + BL + L^T B^T + B_w B_w^T < 0}$$

FREE VARIABLES: Y, L, S

8) Minimization of the H_2 norm [discrete-time]

CONSIDER THE SYSTEM WITH ADDITIVE NOISE w

$$\begin{cases} x_{k+1} = Fx_k + Gu_k + G_w w_k = (F + GK_x) x_k + G_w w_k \\ z_k = Hx_k + Du_k \quad z_u = (H + DK_x) x_k \end{cases}$$

PROBLEM: find K_x such that $u = K_x x$ allows to minimize the H_2 norm of the transfer function $G_{zw}(z)$ from w_k to z_k

Theorem:

$$\|G_{zw}(z)\|_2^2 = \inf \left\{ \text{tr} \left\{ (H+DK_x) P (H+DK_x)^T \right\} \right. \\ \left. \text{s.t. } (F+GK_x) P (F+GK_x)^T - P + G_w G_w^T < 0 \right\}$$

$$\Rightarrow \boxed{\min \text{tr}(S)} \quad , \text{ WHERE } S \text{ FULFILLS (SEE ?)} \quad \downarrow$$

Replace $y \rightarrow P$
 $c \rightarrow H$

$$\Downarrow$$

$$\boxed{\begin{bmatrix} S & HP+DL \\ (HP+DL)^T & P \end{bmatrix} \succeq 0}$$

SUBJECT TO (SEE LMI FOR STABILITY IN DISCRETE TIME)

$$\begin{bmatrix} P - FPF^T - FL^T G^T - GLF^T - G_w G_w^T & GL \\ (GL)^T & P \end{bmatrix} \succeq 0$$

C) DESIGN OF DECENTRALIZED AND DISTRIBUTED CONTROLLERS

GAIN MATRICES:

- CENTRALIZED : $K_x \in \mathbb{R}^{m \times n}$
- DECENTRALIZED : $K_x \in K^d$ BLOCK-DIAGONAL
- DISTRIBUTED : $K_x \in K^N$ SPARSE

To design decentralized and distributed control gains we just have to enforce a structure to gain K_x when formulating the LMI problems!

However, K_x is not free, but it is computed as

$$K_x = L y^{-1} \quad (K_x = L P^{-1})$$

$\Rightarrow$ You can impose a structure to K_x by imposing a structure to L and y (or P).

In particular:

- L must have the same structure of K_x , i.e.,

$$\rightarrow L \in \mathbb{R}^{m \times n}$$

CENTRALIZED CASE

$$\rightarrow L \in K^N$$

DISTRIBUTED CASE

$$\rightarrow L \in K^d$$

DECENTRALIZED CASE

- y (or P) must be block-diagonal

However, this makes the Lyapunov inequalities just SUFFICIENT for stability

EXAMPLE

IF WE CHOSE $L = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$, $P^{-1} = \begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix}$

$$\Rightarrow Kx = LP^{-1} = \begin{bmatrix} 5 & 6 \\ 2 & 1 \end{bmatrix} \Rightarrow \text{IT DOES NOT HAVE THE STRUCTURE OF } L!$$

HOWEVER, IF P IS DIAGONAL

$$P = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix} \Rightarrow Kx = LP^{-1} = \begin{bmatrix} 2 & 5 \\ 2 & 5 \end{bmatrix} \Rightarrow Kx \text{ IS STRUCTURED!}$$

REMARK: NUMBER OF FREE VARIABLES AND COMPUTATIONAL EFFORT

• P block-diagonal, i.e., $P = \begin{bmatrix} P_1 & & 0 \\ & P_2 & \\ 0 & & \ddots & P_n \end{bmatrix}$

• $P_i \in \mathbb{R}^{n_i \times n_i}$

FOR EACH SUBMATRIX P_i THE
NUMBER OF FREE VARIABLES IS
 $\frac{n_i(n_i+1)}{2}$

• L structured $\Rightarrow$ IN THE DECENTRALIZED CASE $L \in K^d$

$$L_i \in K^d = \begin{bmatrix} L_{i1} & L_{i2} & \dots & 0 \\ 0 & L_{i2} & \dots & L_{in} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & L_{in} \end{bmatrix}$$

$$L_{ij} \in \mathbb{R}^{m_i \times n_j} \quad \forall i$$

FOR EACH SUBMATRIX L_{ij} THE NUMBER
OF FREE VARIABLES IS $n_j \times m_i$

$\Rightarrow$ IN THE DISTRIBUTED CASE $L \in K^N$

$$L_{ij} \in K^d = \begin{bmatrix} L_{i1} & L_{i2} & \dots & L_{in} \\ L_{j1} & L_{j2} & \dots & L_{jn} \\ \vdots & \vdots & \ddots & \vdots \end{bmatrix}$$

$$L_{ij} \in \mathbb{R}^{m_i \times n_j} \quad \forall i, j$$

FOR EACH SUBMATRIX L_{ij} THE NUMBER
OF FREE VARIABLES IS $m_i \times \sum_{j \in N_i} n_j$

OVERALL :

$$\mu_{DEC} = \sum_{i=1}^N \left(n_i \frac{(u_i + 1)}{2} + m_i \cdot u_i \right)$$

$$\mu_{DISTR} = \sum_{i=1}^N \left(\frac{n_i (u_i + 1)}{2} + m_i \sum_{j \in N_i} n_j \right)$$

$\Rightarrow$ LESS DEGREES OF FREEDOM THAN IN THE
CENTRALIZED CASE



Less computational effort !!